

SHARAFAT HOSSAIN

Department of Computer and Information Science
University of Arkansas at Little Rock (UALR) Little Rock, Arkansas 72204

shossain1@ualr.edu | 609-255-1641
linkedin.com/in/hossain-sharafat

EDUCATION

University of Arkansas at Little Rock (UALR)	Little Rock, AR
PhD in Computer and Information Science	Expected Jul 2026
Thesis: "Vision-GIS Cross-Attention Transformer-driven Flooded Road Vehicle Dynamics Modelling."	
Advisor: Dr. Serhan Dagtas	
University of Arkansas at Little Rock (UALR)	Little Rock, AR
MSc in Information Science	Dec 2023
Khulna University of Engineering and Technology	Khulna, Bangladesh
BSc in Electronics and Communication Engineering	Oct 2017

RESEARCH INTEREST

- Leveraging hardware-neural architecture to discover compact, edge-optimized vision models with strict VSS latency budgets, maintaining state-of-the-art detection and performance.
- Designing compression-aware, edge-deployable diagnostic models for point-of-care imaging systems, enabling real-time lesion detection and segmentation in resource-limited clinical environments without compromising diagnostic sensitivity

TEACHING EXPERIENCE

University of Arkansas at Little Rock	Little Rock, AR
Guest Lecturer	Spring, 2026
▪ To share the current trends in computer vision, delivered a lecture on my research for the Advanced Data Science & Technology course (30 students). ▪ Developed one-hour lecture with interactive components, including clicker questions and think-pair-share.	
Graduate Teaching Assistant (GTA)	Fall 2024 – Present
▪ Delivering practical and tutorial sessions for courses in Image Processing and Computer Systems for groups of up to 30 undergraduate students ▪ Responsible for prior assessment of curriculum, development of suitable resources, and delivery of sessions and coursework, marking, and feedback. ▪ Administered exams and evaluated assignments, maintaining rigorous and equitable grading standards. ▪ Courses: Data Science & Technology, Image Processing, Database Systems / Info Architecture, Computer Systems.	
Tutor Learning Commons Ottenheimer Library	Spring 2024, Fall 2025
▪ Provided academic support across fundamental Computer Science courses through structured individual and small group sessions. ▪ Supported exam preparation through focused review sessions targeting high-yield topics, common misconceptions, and problem-solving patterns frequently tested in course assessments.	

WORK EXPERIENCE

Avangard AI Inc.

Sacramento, CA (Remote)

AI/ML Engineer Intern

Jan 2026 – Present

- Deployed research models into secure, production-grade ML systems on Google Cloud Platform (GCP).
- Fine-tuned LLMs for domain-specific applications, ensuring alignment with product requirements.
- Monitored experiments and identified model drift in real time using MLflow and Weights & Biases.
- Collaborated with cross-functional teams for testing, documentation, and cloud pipeline improvement.

University of Arkansas at Little Rock

Little Rock, AR

Graduate Research Assistant (GRA)

Mar 2022 – Present

- Built end-to-end multimodal AI pipelines combining CNN-based classification/regression, GAN-based image enhancement, and sensor data fusion for environmental hazard monitoring.
- Integrated hydrodynamic model (SFINCS, RAC2E) outputs with geospatial and vision analytics to support road-condition assessment, routing optimization, and risk-aware situational awareness in disaster response.
- Explored multi-instance learning, cross-modal attention, and weakly supervised localization to address granularity mismatch between slice-level CT inputs and patient-level labels.

Banglalink Digital Communication

Dhaka, Bangladesh

Data & Platform Service Specialist Engineer

Apr 2018 – Oct 2021

- Deployed anomaly detection models to predict network congestion, cell tower outages, and maintenance.
- Integrated ML-driven insights into CRM and billing systems, translating predictive analytics into operational outcomes.
- Optimized traffic routing by forecasting peak-usage periods and reducing network latency.
- Implemented automated churn prediction systems, enabling early risk identification and improving customer retention.

PUBLICATIONS

- Hossain S., Jui N.N., Hossain M.F. "A Ultra Wide Triple-band Antenna for Biomedical Application." 2019 IEEE CCECE, pp. 1–6.
- Hossain S., Fahim M.A.N., Jui N.N., Tarifuzzaman M. "A Simple Nonlinearity Upgradation Based Method of Enhancing Low-Light Images." 2019 IEEE CCECE, pp. 1–6.
- Hossain S., Jui N.N., Hossain M.F., Morshed K.M. "A Simple Triple-band Antenna for Implantable Biomedical Application." 2017 IEEE R10-HTC, pp. 564–568. [Best Paper Award]
- Fahim M.A.N., Hossain S. "A Simple Way to Create Pointillistic Art from Natural Images." 2017 IEEE CYBCONF, pp. 1–5.
- Hossain S., Fahim M.A.N. "A Simple Way of Image Encryption Using Pixel Shuffling and Pixel Manipulation." 2017 ICCIT, pp. 1–4.
- Hossein Pour Z., Hossain S., Liao H., Dagtas S., Mulburn A.B. (2025). "Predicting Vehicle Speeds on Flooded Roads Using Video-based Observations and Probabilistic Models." [Under Review]
- Odebode A., Milburn A.B., Azucena J., Liao H., Huang X., Hossain S., Dagtas S. "Geolocating Social Media Images for Disaster Response: An Integrated Multimodal Approach." [Under Review]
- Hossain S. et al. "When is 3D Worth It? A Resource–Performance Frontier for CNNs and Transformers in Lung CT." [In Preparation]
- Hossain S. et al. "Navigating Flooded Roads: A Comprehensive Data-Driven Depth-Speed Function from Real-World Observations." [In Preparation]

AWARDS & HONORS

Best Paper Award	2017
IEEE Region 10 Humanitarian Technology Conference (R10-HTC) — selected among all international submissions	
DART Travel Grant Award	2022
University of Arkansas at Little Rock — competitively awarded to support high-impact research dissemination	
Dean's Award for Academic Excellence	2014

LEADERSHIP & ACTIVITIES

Association for Computing Machinery (ACM)	Little Rock, AR
Member, UALR Chapter	2023 – Present
Institute of Electrical and Electronics Engineers (IEEE)	Khulna, Bangladesh
Member, KUET Chapter	2014 – Present
AR INBRE Conference / Lisa Academy Science Fair / Central AR Science & Engineering Fair	Arkansas
Judge, Poster & Science Fair Presentations	2025 – 2026
2nd International Conference on Computational and Cognitive Engineering (TCCE)	
Ad Hoc Reviewer	2020
KUET-Dream — Voluntary Blood Donation Society	Khulna, Bangladesh
Member	2014 – Present
UALR Campus Garden Alliance	Little Rock, AR
Member	2023 – Present

SKILLS & INTERESTS

- Languages: Python, C++, R, Java, MATLAB, Verilog, HTML5
- ML & Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Torchvision, Detectron2, MMDetection
- Computer Vision: CNNs, ViT, Swin Transformer, CLIP, DINOv2, SAM/SAM2, Grounding DINO, Florence-2, Qwen2.5-VL, YOLOv5/v8/11/12, RF-DETR, RT-DETR, SegFormer, DeepLabV3+, U-Net, ByteTrack, DeepSort, EDSR, DnCNN, Optical Flow.
- LLM & GenAI: Hugging Face (Transformers, Diffusers, PEFT, TRL), LangChain.
- Vector Databases: FAISS, Pinecone, Qdrant, Milvus, ChromaDB, Weaviate
- Cloud & Big Data: AWS (S3, EC2), GCP, Azure, HPC, Hadoop, Spark, Hive, HBase
- MLOps & Databases: MLflow, W&B, DVC, Docker, MySQL, MongoDB, Elasticsearch, Redis, Tableau
- Networking & Security: TCP/IP, TLS/SSL, SSH, VPN, DNS, OSI Model, WDM, Encryption
- Version Control & DevOps: Git, GitHub, DVC